Anomaly Detection in Autonomous Driving Agents Using World Models

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***Abstract*—End-to-end reinforcement learning agents for autonomous driving remain unreliable when confronted with situations outside their training distribution, and because their perception and planning are optimized jointly, they offer no natural interface at which an independent safety check can be inserted. This paper proposes and evaluates a dual-process anomaly detection architecture for world-model-based driving agents, inspired by the System 1 / System 2 distinction of dual-process cognitive theory. A DreamerV3 agent is trained to follow waypoints in a custom Gymnasium-compatible environment built on the CARLA simulator, using semantically segmented camera observations, and two complementary detectors are then run in real time on the frozen agent. The System 1 detector, a convolutional binary classifier over paired real and imagined (decoder-reconstructed) semantic frames, provides fast local obstacle detection from discrepancies between observed and reconstructed semantic frames. The System 2 monitor, a theoretically grounded Kullback-Leibler novelty threshold computed directly from the Recurrent State-Space Model, continuously assesses whether the agent's world model remains valid in the current environment. Run concurrently on identical data across six evaluation conditions, the two detectors are shown to capture complementary properties of the same experience, local obstacle presence and global distribution shift, and to reach their operating regimes under opposite world-model conditions. This complementarity defines a unified two-tier Anomaly Observation System in which the System 2 monitor modulates the trust placed in the System 1 detector when sustained global novelty indicates degradation of the underlying world model. The system is implemented as a closed-loop safety supervisor that brings the vehicle progressively to a stop while an obstacle remains detected, and it is evaluated against the open-loop agent and against each tier in isolation, over 100 episodes per configuration. With a single vehicle obstacle the supervisor removes every collision in all three driving scenarios, against 75 to 98 collisions in 100 episodes without supervision, and in the well-fitted scenario it stops at a median of 18 m from the obstacle; as the world model's reconstruction degrades, a growing share of the avoided collisions comes from stops far from the obstacle. The combined supervisor preserves the collision avoidance achieved by System 1 in the single-obstacle setting, while the System 2 gate mainly affects behavior under the degraded-world-model condition. Under non-stationary training the reconstruction losses rise while the reward loss keeps decreasing, which suggests that reconstruction-based safety supervision can lose reliability before the control policy fails.**

***Index Terms*—anomaly detection, out-of-distribution detection, world models, DreamerV3, CARLA, autonomous driving, active perception, dual-process theory, semantic segmentation**

I. Introduction

Deploying autonomous vehicles (AVs) in the open world is one of the most demanding challenges of modern engineering. A fully autonomous vehicle must perceive complex, dynamic scenes with high fidelity, plan safe trajectories, and execute control in real time, while maintaining safety guarantees that extend beyond the nominal operating conditions encountered during training [1, 2]. Although substantial progress has been made on individual sub-problems such as object detection, motion planning, and sensor fusion, integrating these capabilities into a robust, generalizable system remains open.

The core difficulty is distributional robustness. A deep network trained on a finite set of driving scenarios will inevitably face unfamiliar situations at deployment time, unexpected road users, extreme weather, construction zones, or mechanical faults. In classical modular pipelines, such out-of-distribution (OOD) events can be handled at the perception stage by dedicated detection mechanisms. In end-to-end learned systems, perception and planning are optimized jointly; failure states become less legible, and the need for an independent safety supervision layer becomes more acute [2, 3].

World-model-based reinforcement learning (RL), exemplified by DreamerV3 [4], is a natural substrate for such supervision. The agent learns a compressed latent model of the world and optimizes its policy entirely inside imagined rollouts. The internal world model is a probabilistic model of the environment's transition dynamics, and the central quantity explored in this work is the divergence between the agent's imagined internal state and the reality it perceives through its sensors. This divergence is exploited to detect anomalies, i.e., states outside the training distribution.

A. A Dual-Process View of Anomaly Detection

Dual-process theories of cognition [5] distinguish between System 1 (fast, automatic, pattern-driven, spatially grounded perception) and System 2 (slow, deliberative, global monitoring of whether one's current model of the situation is still valid).

We argue that anomaly detection in active perception naturally decomposes along exactly this axis, and we instantiate both processes on top of a single world model:

- **System 1: local, spatial, reactive.** A supervised convolutional neural network (CNN) binary classifier compares, at every timestep, the real semantically segmented observation with the world model's imagined reconstruction of the same scene. An unexpected object manifests as a localized class disagreement (e.g., a car-labeled region present in reality but absent from imagination). The classifier answers the fast perceptual question: "Is there an unexpected object in my path?"
- **System 2: global, distributional, metacognitive.** A theoretically grounded KL-divergence threshold, following Zollicoffer et al. [7], is computed directly from the RSSM's prior and posterior latent distributions. It summarizes the model's "surprise" over the entire latent space into a single scalar and answers the deliberative question: "Am I in an environment my world model still understands?"

Our central empirical finding is that these two signals capture complementary properties of the same experience: the classifier responds to local obstacle presence and the KL monitor to global distribution shift, and each reaches its operating regime under the world-model conditions in which the other saturates. The System 1 classifier is at its best precisely when the world model is well fitted, exactly when the System 2 monitor is silent, whereas the System 2 monitor fires aggressively precisely when the world model has broken down, exactly when the classifier's precision is reduced. This complementary regime-dependent behavior forms the basis of the two-tier Anomaly Observation System of Section VI. The System 1 / System 2 terminology is used here as a functional analogy rather than as a cognitive model: System 1 denotes the fast local detector, whereas System 2 denotes the slower, temporally aggregated monitor of global world-model validity.

B. Contributions

The main contributions are:

1. A dual-tier safety-monitoring framework for world-model-based driving agents that combines local reconstruction-based obstacle detection with latent distribution-shift monitoring.
2. An empirical characterization of the different operating regimes of these two signals under increasingly challenging world-model conditions.
3. A closed-loop supervisor in which a temporally filtered world-model novelty signal modulates the trust placed in the local obstacle detector before progressive braking is applied.
4. A CARLA/DreamerV3 implementation and evaluation across static-route, random-route, and random-town scenarios, including both single-vehicle and heterogeneous static-obstacle conditions.

The implementation, including a Gymnasium-compatible CARLA environment and an online DreamerV3 implementation of the KL novelty rule of [7], is publicly available at `https://github.com/kostasoiko4/Anomaly-Detection-in-Autonomous-Driving-Agents-Using-World-Models`.

II. Related Work

Active perception. Bajcsy [8] framed intelligent perception not as passive registration but as a goal-directed control loop in which the agent actively selects sensing actions to minimize uncertainty about the world state; the framework was revisited three decades later [9], and has since been extended to practical AV settings, e.g., uncertainty-guided light-curtain sensing for 3D detection [10]. The DreamerV3 world model can be viewed as active perception in latent space: an internal generative model used to actively predict future observations, closing the action-perception loop. Anomaly detection then corresponds, within this framework, to identifying when the current belief state is no longer a reliable guide for action. The proposed supervision mechanism is related to active-perception principles in that internal predictions are continuously compared with incoming sensory evidence and the resulting uncertainty influences subsequent control. Unlike classical active-perception methods, however, the vehicle does not actively reposition a sensor or select a sensing modality; the intervention considered here is a safety-control action.

Autonomous driving architectures. Modular pipelines decompose driving into perception, prediction, planning, and control [11, 12], with increasing sensor heterogeneity at higher automation levels [13, 14]. End-to-end learning addresses the brittleness and joint-optimization difficulties of modular stacks but makes failure states less distinct, motivating independent OOD supervision [1–3]. Reconstruction-based and distribution-based OOD detection form the two dominant families [2, 15]; the present work sits at their intersection, exploiting the RSSM's probabilistic structure (distribution-based) while using image reconstruction as the reference signal for a discriminative classifier (reconstruction-based).

World models for driving. CarDreamer [16] provides an open platform for world-model-based driving research on CARLA [17], and reports design findings adopted here: speed-based rewards outperform position-based rewards, and semantic observations with explicit intention signals (planned waypoints) improve safety. Synthetic data has been shown to approximate real datasets statistically [18], supporting simulator-based evaluation.

Novelty detection with world models. Zollicoffer et al. [7] define novelty as a persistent inference-time change in the observation space or transition dynamics that invalidates the trained policy, and derive a variable detection threshold from three KL terms computed on the RSSM. Our work provides a real-time DreamerV3-internal implementation of this method (prior implementations targeted DreamerV2) in a closed-loop driving simulator with injected obstacles, and characterizes its operational scope by running it concurrently, on identical data, with a discriminative local detector that targets a complementary signal.

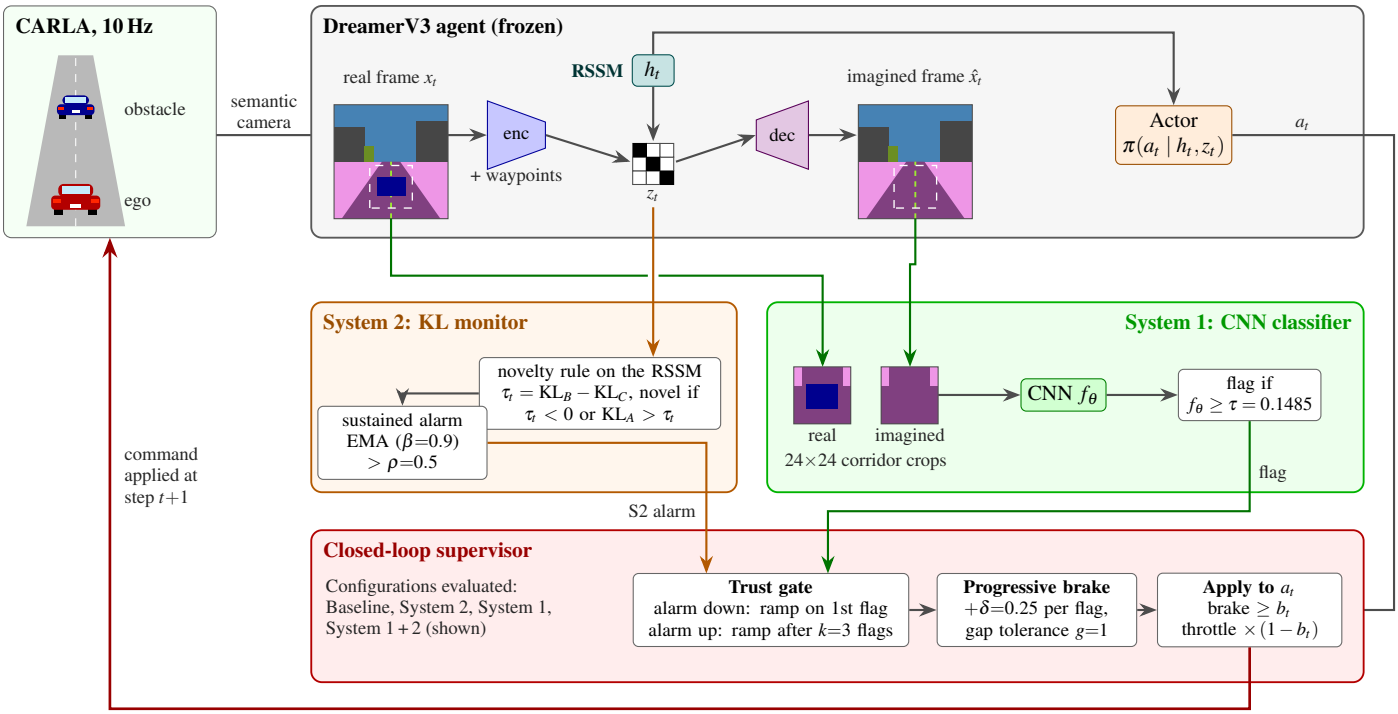


Figure 1. Architecture of the Anomaly Observation System. The semantic camera captures the obstacle; the frozen DreamerV3 agent encodes the frame into its RSSM state, from which the actor selects the driving action and the decoder reconstructs the imagined frame, the scene as the world model expects it. System 1 compares 24×24 class-index crops of the drivable corridor (dashed window) in the real and imagined frames and flags a local obstacle. System 2 evaluates the KL novelty rule on the RSSM distributions and raises an alarm when novelty is sustained. The supervisor turns runs of System 1 flags into a progressive brake, starts the ramp later while the System 2 alarm is up, and applies the brake to the actor's command at the next step.

III. Background

A. DreamerV3 and the RSSM

DreamerV3 [4, 19] learns a world model from high-dimensional observations and optimizes an actor-critic pair purely in imagination. The world model is a Recurrent State-Space Model (RSSM) that factorizes the latent state into a deterministic component h_t, produced by a GRU,

$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1}), \tag{1}$$

and a stochastic component z_t with a posterior (representation) model $q_\phi(z_t \mid h_t, x_t)$ conditioned on the current observation, and a prior (transition) model $p_\phi(z_t \mid h_t)$ conditioned on history alone. Decoders reconstruct the observation, reward, and episode-continuation flag from (h_t, z_t). DreamerV3 uses categorical latents (32 groups of 32 classes, a 1024-dimensional representation), symlog transforms, two-hot reward encoding, and free-bits KL regularization [4]. The training KL term $\mathrm{KL}[q_\phi(z_t \mid h_t, x_t) \,\|\, p_\phi(z_t \mid h_t)]$ encourages the prior to predict the posterior; the same quantity, repurposed at inference time, is the raw material of the System 2 monitor.

B. KL Novelty Threshold

Following [7], let h_0 denote a zero (empty-history) recurrent state. Three KL terms are computed at every step, all with the full posterior as the base distribution:

$$\mathrm{KL}_A = \mathrm{KL}\left[q_\phi(z_t \mid h_t, x_t) \,\|\, p_\phi(z_t \mid h_t)\right], \tag{2}$$

$$\mathrm{KL}_B = \mathrm{KL}\left[q_\phi(z_t \mid h_t, x_t) \,\|\, p_\phi(z_t \mid h_0)\right], \tag{3}$$

$$\mathrm{KL}_C = \mathrm{KL}\left[q_\phi(z_t \mid h_t, x_t) \,\|\, q_\phi(z_t \mid h_0, x_t)\right]. \tag{4}$$

KL_A is the standard world-model surprise; KL_B measures divergence from a history-free prior; KL_C isolates the contribution of history to the posterior by feeding the representation model an empty history but the true observation. The dynamic threshold and decision rule are

$$\tau_t = \mathrm{KL}_B - \mathrm{KL}_C, \qquad \text{novel} \iff (\tau_t < 0) \vee (\mathrm{KL}_A > \tau_t). \tag{5}$$

Since $\mathrm{KL}_A \geq 0$ by definition, a negative threshold satisfies the disjunction automatically, so the rule fires unconditionally in that case. Intuitively, if the current observation is more surprising given the history than it would be without any history, the history provides misleading context, a hallmark of an anomaly that has invalidated the world model.

IV. Problem Formulation and Methodology

This section develops the proposed dual-process architecture at a formal level: the driving task is cast as a partially observable Markov decision process (POMDP), the training curricula are characterized as distributions over environment configurations, and the two detectors are defined as measurable decision

functions of the agent's information state. Implementation-level detail is deferred to Section V-A. Fig. 1 summarizes the complete pipeline, from the obstacle in the simulator to the brake command of the supervisor of Section VI.

A. The Driving Task as a POMDP

The interaction between agent and simulator is modeled as a POMDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma, \Omega, \mathcal{E})$ [6], where $\mathcal{S}$ is the (hidden) simulator state space, $\mathcal{A}$ the action space, $\mathcal{P}(s' \mid s, a)$ the transition kernel, R the reward function, $\gamma \in [0,1)$ the discount factor, Ω the observation space, and $\mathcal{E}(o \mid s)$ the emission function. Time is discretized with a fixed step Δt under synchronous simulation [17], so that $\mathcal{P}$ is well defined and episode evolution is reproducible.

Observation space. The agent observes $o_t = (x_t, v_t) \in \Omega = \mathcal{X} \times \mathbb{R}^{d_v}$. The exteroceptive channel $x_t \in \mathcal{X}$ is a semantically segmented image: the emission factors as $\mathcal{E} = c \circ \ell \circ \pi_{\text{cam}}$, where π_{cam} is the perspective projection of the scene onto the image grid Λ (a roof-mounted camera with fixed extrinsic pose and horizontal field of view), $\ell : \Lambda \to \mathcal{K}$ assigns to every pixel one of K semantic classes from the CityScapes palette [20], and $c : \mathcal{K} \to [0,255]^3$ is an injective palette encoding. The essential property of this channel is nuisance invariance: for the group of photometric nuisance transformations g (illumination, weather, reflectance changes), the labeling satisfies $\ell(g \cdot s) = \ell(s)$, i.e., the observation is constant on nuisance orbits. This both stabilizes world-model learning and sharpens the discrepancy signal between observed and imagined scenes, since any residual difference must be semantic rather than photometric.

The proprioceptive channel $v_t \in \mathbb{R}^{d_v}$ comprises the next n route waypoints expressed in the vehicle frame, i.e., the images of the world-frame waypoints under the rigid transform $T_t \in SE(2)$ from world to ego coordinates, each as a (longitudinal, lateral) pair, together with the scalar speed and the heading deviation from the next waypoint, giving $d_v = 2n+2$. Expressing the route information in the vehicle frame removes dependence on absolute waypoint coordinates and facilitates reuse of the same policy representation across different routes and map locations.

Action space. The action space is the finite Cartesian product $\mathcal{A} = \mathcal{A}_a \times \mathcal{A}_\delta$ of a small set of longitudinal accelerations $\mathcal{A}_a$ and steering values $\mathcal{A}_\delta$ [16]. A fixed decoding map $u : \mathcal{A} \to [0,1]^2 \times [-1,1]$ sends the positive and negative parts of the commanded acceleration, normalized by the maximum commanded magnitude, to throttle and brake, respectively, together with the steering angle. Discretization admits a categorical policy head, matching DreamerV3's discrete-latent design.

Reward and termination. The reward decomposes additively into shaping terms,

$$R = w_{\text{wp}} r_{\text{wp}} + w_{\text{spd}} r_{\text{spd}} + w_{\text{col}} r_{\text{col}} + w_{\text{ln}} r_{\text{ln}} + w_{\text{dst}} r_{\text{dst}} - w_t, \quad (6)$$

where the speed term rewards the velocity component toward the next waypoint while penalizing lateral slip and deviation from a desired speed,

$$r_{\text{spd}} = v_{\text{des}} - \left| v_{\parallel} - v_{\text{des}} \right| - \lambda \min(v_{\perp}, c), \quad (7)$$

with λ a slip-penalty weight and c a cap on the penalized lateral speed; the waypoint term issues a unit reward when the distance to the next waypoint falls below an absorption radius and a terminal penalty when it exceeds a divergence radius; r_{col} and r_{ln} are indicator-based penalties for collision and solid-lane violation (the latter scaled by lateral excursion; dashed markings are not penalized); r_{dst} is a terminal completion bonus; and $-w_t$ is a constant time penalty discouraging inefficient driving. Speed-based rather than position-based shaping is adopted following the empirical finding of [16] that position-only rewards admit a degenerate stationary optimum. The terminal set consists of collision, solid-lane violation or road departure, waypoint divergence, horizon exhaustion, and goal attainment.

B. World-Model Learning and Training Curricula

The world-model parameters ϕ are learned by minimizing the composite objective of DreamerV3 [4] over replayed trajectory segments,

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}\Big[\sum_t \beta_{\text{pred}} \mathcal{L}_{\text{pred}} + \beta_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \beta_{\text{rep}} \mathcal{L}_{\text{rep}} \Big], \quad (8)$$

with the prediction loss $\mathcal{L}_{\text{pred}} = -\ln p_\phi(x_t \mid z_t, h_t) - \ln p_\phi(r_t \mid z_t, h_t) - \ln p_\phi(c_t \mid z_t, h_t)$ and the free-bits-regularized divergence terms

$$\mathcal{L}_{\text{dyn}} = \max\big(1, \mathrm{KL}[\mathrm{sg}(q_\phi(z_t \mid h_t, x_t)) \,\|\, p_\phi(z_t \mid h_t)]\big), \quad (9)$$

$$\mathcal{L}_{\text{rep}} = \max\big(1, \mathrm{KL}[q_\phi(z_t \mid h_t, x_t) \,\|\, \mathrm{sg}(p_\phi(z_t \mid h_t))]\big), \quad (10)$$

where $\mathrm{sg}(\cdot)$ denotes the stop-gradient operator and $(\beta_{\text{pred}}, \beta_{\text{dyn}}, \beta_{\text{rep}}) = (1, 0.5, 0.1)$, the SheepRL defaults used in this study. The actor and critic are optimized purely on imagined rollouts of the prior, as in [4].

Each training scenario is characterized as a distribution μ over environment configurations $\theta = (m, \rho)$, where m is the town map and ρ the route:

- **S1 (Static route):** $\mu_1 = \delta_{(m_0, \rho_0)}$, a Dirac measure on a fixed map and route; the data-generating process is stationary and maximally narrow.
- **S2 (Random route):** $\mu_2 = \delta_{m_0} \otimes \mathrm{Unif}(\mathcal{R}_{m_0})$, with routes drawn uniformly per episode from the feasible route set of the fixed map; stationary but broader.
- **S3 (Random towns):** the map itself is resampled at a preset episode interval and the route per episode, rendering the data-generating process non-stationary on the timescale of the replay buffer.

The non-stationarity of S3 introduces an additional challenge for world-model learning: the replay buffer contains trajectory segments originating from changing emission distributions, which may hinder stable reconstruction learning; empirically, the S3 reconstruction losses do not converge (Section V-B). Each scenario is trained from independent random initialization, without weight transfer, to preserve experimental independence.

C. System 1: Discriminative Local Anomaly Detection

Imagination as a nominal reference. At evaluation time the agent is frozen; let $\hat{x}_t$ denote the mean of the observation

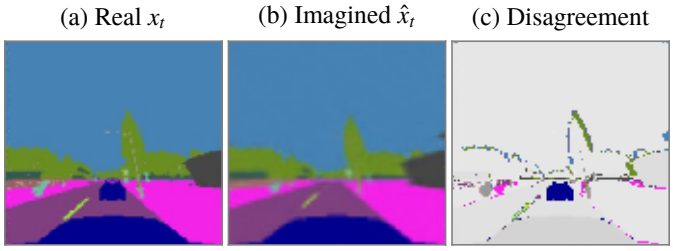


Figure 2. Real observation and imagined frame at the same timestep. (a) The semantic camera frame x_t contains a parked car on the route. (b) The decoder reconstruction $\hat{x}_t$ reproduces the road, vegetation and buildings but not the car. (c) Pixels whose semantic class differs between (a) and (b), shown in the class colour of the real frame. Apart from thin bands along class boundaries, which the decoder blurs, the obstacle is the only large compact region of disagreement; this is the discrepancy System 1 classifies within the corridor crop.

decoder $p_\phi(\hat{x}_t \mid h_t, z_t)$ evaluated at the full posterior latent. Since the training distribution contains no obstacles, the decoder is a mapping onto (an approximation of) the manifold of nominal scenes: even though the posterior z_t has encoded the current observation, the reconstruction reproduces the world as learned. An injected obstacle therefore manifests as a residual component of x_t that is, by construction, absent from $\hat{x}_t$, a spatially localized semantic discrepancy (Fig. 2).

Preprocessing as operators. Two operators map images into the detector's input space. The quantization operator $Q : \mathcal{X} \to \mathcal{K}^\Lambda$ recovers the class-index field by nearest-neighbor inversion of the palette encoding,

$$Q(x)[i,j] = \arg\min_{k \in \mathcal{K}} \|x[i,j] - c(k)\|_2^2, \tag{11}$$

and the restriction operator ρ_W crops the field to a fixed window $W \subset \Lambda$ covering the drivable corridor, retaining the information-rich portion of the scene while reducing input dimensionality. The detector input is the pair

$$u_t = \big(\rho_W Q(x_t),\ \rho_W Q(\hat{x}_t)\big) \in \mathcal{K}^{|W|} \times \mathcal{K}^{|W|}. \tag{12}$$

Hypothesis class and learning objective. The detector is a parametric function $f_\theta : \mathcal{K}^{|W|} \times \mathcal{K}^{|W|} \to (0,1)$ drawn from the class of convolutional networks: compositions of discrete convolutions (encoding local connectivity and translation equivariance [25]), pointwise nonlinearities, normalization [24], and spatial max-pooling (conferring local translation invariance), terminated by an affine map and a logistic link. Training minimizes the empirical binary cross-entropy. The sigmoid output is used as a continuous anomaly score in $(0,1)$, which is converted into a binary decision through a threshold selected on the validation set, $\hat{y}_t = \mathbf{1}\{f_\theta(u_t) \geq \tau\}$. The threshold serves a single purpose: it turns the continuous sigmoid output into the binary flag that the safety supervisor consumes. Under the episode-level split of Section V-A the score distribution is concentrated near $\{0,1\}$ but is not bipolar: a fraction of the outputs lies in the intermediate zone and a minority of anomalous frames receives scores near zero, so the choice of τ is an operating-point decision. It is made on the validation split from the receiver operating characteristic of the sigmoid output (Section V-A, Fig. 10), as the largest τ that retains a recall floor; the safety-conservative direction, preferring false alarms to missed detections, is thus expressed as the recall floor, while the suppression of transient false alarms is handed to the temporal logic of the supervisor (Section VI).

Supervision signal. The per-timestep ground-truth label $y_t \in \{0,1\}$ is defined by a morphological region-comparison operator on the pair of class-index fields, with one variant per obstacle regime. In the single-obstacle regime the operator is class-conditional: an obstacle is a connected component of the class car in $Q(x_t)$ whose overlap with the car support of $Q(\hat{x}_t)$ falls below a coverage ratio of its area. In the random-obstacle regime the operator is class-agnostic: a drivable corridor is formed as the dilated union of the road supports of both fields, object pixels are the non-road, non-sky pixels of each field, and an obstacle is a connected component of camera object pixels inside the corridor that the (dilated) imagined object support leaves uncovered, subject to a minimum-area constraint. The operator is fully automated: it labels the frames of the classifier's training dataset offline, and the same operator produces the per-timestep ground truth online during the comparative evaluation of Section IV-E, so that no frame is labeled by hand. A label of 1 marks frames in which the operator finds an object in the corridor that the reconstruction does not contain, which occurs mainly while the obstacle is in view; episodes are therefore anomalous mainly over a suffix of their duration, yielding an inherently imbalanced label distribution. The regime is selected by a single configuration switch shared by the environment and the evaluation, so that the obstacles spawned and the labels produced always refer to the same regime.

D. System 2: Latent Distribution-Shift Monitoring

The System 2 monitor evaluates the decision rule of (5) at every timestep. All three divergences (2)–(4) share the full posterior as base distribution and differ only in the reference. Because the stochastic latent factorizes as a product of G independent categorical variables, each divergence decomposes additively over factors, $\mathrm{KL} = \sum_{g=1}^{G} \mathrm{KL}_g$, by the additivity of relative entropy over independent coordinates. Computationally, the monitor requires per step two additional prior evaluations, at h_t and at the empty history $h_0 = 0$, one additional posterior evaluation at (h_0, x_t), and one re-encoding of the current observation that feeds the empty-history posterior; the recurrent state h_t and the full posterior $q_\phi(z_t \mid h_t, x_t)$ are byproducts of action selection, so the monitor adds a small constant overhead and no auxiliary recurrent rollout.

Conceptually, the monitor performs a per-step hypothesis test on model validity: under the null hypothesis of a well-fitted world model, history is informative and KL_A remains small relative to $\tau_t = \mathrm{KL}_B - \mathrm{KL}_C$; a violation of the rule indicates that the learned history either misleads the prediction ($\tau_t < 0$) or fails to explain the observation better than ignorance would ($\mathrm{KL}_A > \tau_t$).

E. Evaluation Protocol

Both detectors are deterministic functions of the information available to the agent at time t, the observation-action history

up to o_t. The evaluation proceeds in two passes over the same frozen agent.

Open-loop detection pass. The detectors run concurrently and independently, each computing its per-step decision, $D_t^{(1)}$ (System 1) and $D_t^{(2)}$ (System 2), from its own signal on the same frame; both decisions are recorded against the common obstacle label y_t. This shared label should be read asymmetrically: for System 1 it measures the task the detector was trained for (local obstacle presence), whereas for System 2 it serves as a probe that delineates the operational scope of a monitor designed for a different task (global model validity). Detector outputs are recorded and never fed back into control in this pass, which isolates detection quality from policy-interaction effects. Because anomaly prevalence is low (Table I), evaluation emphasizes prevalence-robust statistics: balanced accuracy (the mean of recall and specificity) and the Matthews correlation coefficient,

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \tag{13}$$

reported alongside accuracy, precision, recall, specificity, F1, and error rates.

Closed-loop safety pass. The two detector bits are fed to the Anomaly Observation System of Section VI, whose command is applied to the vehicle on the following step (one control period of 0.1 s). The actuator is the same in every intervening configuration, a brake that ramps to a full stop while the detection persists, so that differences in outcome are attributable to the detection logic. The pass is run under four supervisor configurations, the system under test and its ablations: BASELINE (the open-loop agent, identical to the detection pass), SYSTEM 1 (the classifier alone brakes), SYSTEM 2 (the KL monitor alone brakes on a sustained alarm) and SYSTEM 1 + 2 (the two-tier Anomaly Observation System). Each configuration is a separate run of 100 episodes. Routes, obstacle types and obstacle placements are drawn independently in every run, so that each configuration faces its own sample of events and the comparison between configurations rests on their confidence intervals.

An episode ends in one of six outcomes: collision, safe stop (the supervisor has brought the vehicle to rest and the episode is truncated after a fixed number of stationary steps), lane departure, route completion, waypoint loss, or timeout. The closed-loop metrics are defined on the ego-obstacle distance, which the simulator provides at every step, so that they are independent of the labeling operator. A safe stop that ends within 25 m of the obstacle is an *obstacle stop*, one that ends farther from it an *early stop*. Per configuration we report the collision rate with its 95% Wilson interval, the obstacle-stop and early-stop rates, the stopping margin (ego-obstacle distance at an obstacle stop), the brake-onset distance (ego-obstacle distance at the first frame of the brake run that ends in an obstacle stop), the impact speed on the frame preceding a collision, the fraction of frames braked while the obstacle is more than 25 m away (early braking), and the fraction of frames on which the System 2 alarm is up. The detection metrics of Section V-C are taken from the open-loop pass.

V. EXPERIMENTAL RESULTS

Two obstacle regimes are evaluated: R1 (a single static vehicle of class car placed at a random route point) and R2 (a randomly drawn obstacle type per episode), each crossed with the three driving scenarios S1–S3, yielding six conditions. Tables II and III report the full metric sets for the two detectors side by side on identical data; following the protocol of Section IV-E, these joint tables characterize the two complementary signals of the dual-process architecture against one probe label. The corresponding confusion matrices and summary plots are given in Figs. 5–9.

A. Experimental Setup and Implementation

Environment and agent. The environment of Section IV-A is realized as a Gymnasium-compatible [21, 22] interface to CARLA 0.9.15 [17] (ego vehicle: Tesla Model 3; routes generated by the simulator's global route planner, adapted to emit dense waypoint sequences), running synchronously at $\Delta t = 0.1$ s (10 Hz). The semantic camera is roof-mounted at height 1.6 m with pitch 30° and a 120° horizontal field of view, and produces $128 \times 128 \times 3$ palette-encoded frames over $K = 19$ CityScapes classes, downsampled internally by the world model to 64×64. The state vector uses the next $n = 5$ waypoints ($d_v = 12$); the action grid instantiates $\mathcal{A}_a = \{-2, 0, 2\}$ m/s^2 and $\mathcal{A}_\delta = \{-0.6, -0.2, 0, 0.2, 0.6\}$, so $|\mathcal{A}| = 15$, with throttle/brake normalization $u_{\mathrm{thr}} = \min(a^+/3, 1)$, $u_{\mathrm{brk}} = \min(a^-/3, 1)$. The waypoint absorption and divergence radii are 1.8 m and 15 m, the speed term of (7) uses $(\lambda, c) = (2, 0.5)$, and in S3 the town map is resampled every 20 episodes. The quantization and restriction operators of Section IV-C act on the 64×64 fields with a fixed 24×24 window over the drivable corridor. The class-conditional labeling operator uses a coverage ratio of 0.3; the class-agnostic operator dilates the road corridor by two pixels, the imagined object support by one pixel, and keeps components of at least three pixels.

World model and detector implementation. DreamerV3 is trained with the SheepRL framework [23]: batch size 16, sequence length 64, replay ratio 1, replay buffer 10^5; a convolutional encoder/decoder pair for the semantic image and a multilayer-perceptron encoder for the $d_v = 12$-dimensional state vector. The System 1 detector instantiates the hypothesis class of Section IV-C as a compact two-block convolutional network (64 filters of size 5×5, then 64 of size 3×3, each followed by batch normalization and 2×2 max-pooling, with dropout of $p = 0.5$, a 128-unit hidden layer, and a logistic output), optimized with Adam (learning rate 10^{-3}) under binary cross-entropy.

Runtime was measured on an Intel Core i7-11700K CPU and an NVIDIA GeForce RTX 3080 GPU, over 10 episodes per scenario of the SYSTEM 1 + 2 configuration. Per frame, DreamerV3 action selection required 2.5 ms on average, the world-model pass that computes the System 2 divergences and

the reconstruction 5.9–6.7 ms, System 1 (preprocessing and classifier) 24.0–24.8 ms, and the supervisor 0.06 ms. The complete monitoring pipeline took 33.2–33.3 ms on average, with a 99th percentile of at most 42.9 ms, and remained below the simulator control period of 100 ms at 10 Hz on 99.8–99.9% of the steps; the few exceptions, up to 252 ms, include the first step of each run.

Dataset collection. The experimental pipeline proceeds in three phases. In the first phase, once trained, the agent is frozen and run in the open-loop configuration with injected obstacles of the random-obstacle regime, which contains the single-obstacle case; at every timestep the paired real/imagined frames are stored under the episode and timestep indices that produced them, and each frame is labeled normal or anomalous by the class-agnostic operator of Section IV-C. The driving scenarios S1–S3 parameterize world-model training and the comparative evaluation; a single classifier is trained on this collection and used in both evaluation regimes.

Classifier training and data split. In the second phase the collected frames are partitioned into training, validation and test sets *by whole episodes*. The episode identifiers present in the collection are sorted and cut into three contiguous ranges such that the training, validation and test sets hold 70%, 15% and 15% of the collected frames, up to the granularity of one episode; every frame of an episode, nominal and anomalous alike, belongs to the split of its episode, and the frames are shuffled within each split only. The validation set is monitored during optimization and the test set is evaluated once. Since consecutive frames at 10 Hz are strongly correlated, this partition ensures that a held-out frame never shares an episode with a training frame. The collection yields 17,756 frame pairs over 100 episodes. The split assigns 62 episodes (12,516 frames, 70.5%, of which 19.0% anomalous) to training, 18 episodes (2,601 frames, 14.6%, 7.3% anomalous) to validation and 20 episodes (2,639 frames, 14.9%, 10.0% anomalous) to test. After training, the network reaches a cross-entropy of $1.0 \cdot 10^{-3}$ and accuracy 0.9997 on the training split, 0.323 and 0.947 on the validation split, and 0.246 and 0.966 on the held-out test episodes (accuracies at the default 0.5 cut).

Decision threshold. The threshold τ of Section IV-C is selected on the validation split (Fig. 10). The sigmoid output separates the two classes with a ROC area of 0.956 and an average precision of 0.798; 7.5% of the validation scores fall in $(0.01, 0.99)$. The limit $\tau \to 0^+$ ($\tau = 10^{-6}$) reaches a recall of 0.974 at a false-positive rate of 0.348, i.e. one nominal frame in three is flagged; recall then falls slowly along the curve while the false-positive rate drops steeply, and the curve flattens at a recall of about 0.81. The threshold is set as the largest τ that retains a validation recall of at least 0.81, giving $\tau = 0.149$ (validation recall 0.811, false-positive rate 0.053, precision 0.548). On the held-out test episodes this threshold yields accuracy 0.956, precision 0.714, recall 0.940 and a false-positive rate of 0.042 (249 true positives, 100 false positives, 16 misses over 2,639 frames). The Youden optimum of the same curve, $\tau = 0.026$, recovers three additional anomalous test frames at the cost of 41 additional false positives.

Table I
PER-TIMESTEP SAMPLE COUNTS AND ANOMALY PREVALENCE PER CONDITION IN THE COMPARATIVE-EVALUATION PASS (100 EPISODES EACH).

Regime	Scenario	Samples	Anomalies
R1 Single	S1 Static route	25,185	21.0%
R1 Single	S2 Random route	21,921	23.3%
R1 Single	S3 Random towns	19,046	25.2%
R2 Random	S1 Static route	26,135	2.7%
R2 Random	S2 Random route	22,318	4.6%
R2 Random	S3 Random towns	15,230	11.0%

Non-learned baseline. To test whether learning is needed, the same protocol was applied to the fraction of corridor pixels whose class differs between $\rho_W Q(x_t)$ and $\rho_W Q(\hat{x}_t)$, with the threshold selected by the same rule on the validation split. On the held-out test episodes this mismatch fraction reaches a recall of 0.913 at a false-positive rate of 0.318 (precision 0.243, MCC 0.369, ROC area 0.912); counting only pixels whose observed class is neither road nor sky gives a recall of 0.875 at a false-positive rate of 0.254 (precision 0.278, MCC 0.402, ROC area 0.915). The classifier reaches a recall of 0.940 at a false-positive rate of 0.042 (precision 0.714, MCC 0.796, ROC area 0.990). A disagreement count alone does not separate obstacle residuals from ordinary reconstruction errors; the classifier does so at a far lower false-positive rate.

Comparative evaluation pass. In the third phase, with the classifier trained, the frozen agent is run in a further, separate pass of 100 episodes per condition, during which both detectors operate simultaneously and independently on every frame, the System 1 classifier on the paired class-index maps and the System 2 rule on the RSSM divergences, so that their decisions refer to exactly the same data at exactly the same instants. All detection metrics of Section V-C are computed from this pass. The System 2 divergences are computed on the straight-through categorical latents ($G = 32$ groups of 32 classes), treating the groups as independent factors. The closed-loop safety pass of Section IV-E reuses the same evaluation machinery with the supervisor enabled.

In every episode with obstacles, a static obstacle is spawned at a uniformly random route point at least 10 waypoints after the start and on a straight approach (route heading change of at most 15° over the preceding 15 waypoints), so that each episode begins nominal and the obstacle enters the camera's view before the vehicle reaches it. The random-obstacle pool draws from the simulator's vehicle and static-prop blueprints; pedestrians are excluded since the simulator registers no collision when the ego vehicle pushes one. The agent's recurrent state is reset at the start of every episode, so the history the System 2 monitor conditions on belongs to the current episode alone. Table I lists per-timestep sample counts and anomaly prevalence for the comparative-evaluation pass.

B. World-Model Training Outcomes

The three curricula of Section IV-B produce qualitatively distinct convergence behavior, summarized by the mean-reward and episode-length curves of Fig. 3. S1 converges fully by ≈100k environment steps (mean-reward plateau ≈2000; episode length saturating at the ≈500-step horizon, indicating consistent route completion without premature termination, a fully solved scenario and the ideal baseline for detector evaluation). S2 (∼250k steps) also converges, but with pronounced variance attributable to per-route difficulty: after an initial ascent it settles in a mean-reward band of ∼1200–1500 (episode lengths ∼320–380) that it maintains without collapse from ∼100k steps onward, its losses still trending downward, evidence of a generalizable waypoint-following policy rather than route memorization. S3 (∼550k steps) exhibits a characteristic sawtooth profile in both curves: performance rises within each town and collapses at every 20-episode map change (reward dropping back below ∼200, episode length to ∼50–100 steps), with per-town peaks growing over training (raw peaks below ∼400 in the first 100k steps, ∼1300–1700 after 300k) but never forming a stable plateau.

Critically, S3 never converges in reconstruction: the observation loss rises from ∼11 to ∼17, the state (KL) loss from ∼3 to ∼8, and the total world-model loss from ∼15 to ∼25, consistent with the hypothesis that the changing training distribution of Section IV-B contributes to reconstruction instability, while the reward loss keeps decreasing (Fig. 4).

We refer to this divergence between the reward and reconstruction losses as control-reconstruction decoupling, and use it to interpret detector behavior in Section V-E.

C. System 1 Performance

In the well-fitted scenario S1 the classifier reaches its highest MCC in both regimes (0.572 in R1, 0.615 in R2; Tables II–III). Its figures differ between the two regimes because the two labeling operators fire at different ranges. The class-conditional operator of R1 labels the car from a distance of about 50 m: in R1/S1 it marks 58.6% of the frames 30–50 m from the obstacle and 99.9% of those at 20–30 m. The class-agnostic operator, on which the classifier is trained, fires mostly within 20 m: in R2/S1 it marks 1.1% and 6.9% of the frames in the same two bands. The classifier follows the range of its training label. In R1/S1 its recall is 0.99 at 10–20 m, 0.31 at 20–30 m and 0.06 at 30–50 m, which yields an overall recall of 0.488 at a precision of 0.833 against the class-conditional label, against a recall of 0.941 in R2/S1. The classifier therefore detects obstacles in the near field of the corridor, and the R1 figures additionally count the frames in which a distant car is already visible to the class-conditional operator.

Precision falls as the reference reconstruction degrades: in R1 from 0.833 in S1 to 0.458 in S2 and 0.588 in S3, in R2 from 0.419 to 0.150 and 0.273, with false-positive rates rising from 0.026 and 0.037 in S1 to 0.20 in S2 and to 0.15 and 0.23 in S3. The random-route scenario S2 yields the lowest precision in both regimes, which is consistent with its world-model losses still decreasing at the end of training (Section V-B). The lower precision of R2 in every scenario also reflects prevalence: the class-agnostic label marks 2.7–11.0% of the frames, against 21.0–25.2% in R1 (Table I). Fig. 6 shows where the additional detections arise. Within 20 m of the vehicle obstacle the classifier fires on at least 98.6% of the frames in every scenario; more than 50 m from the obstacle it fires on 2–3% of the frames in S1, against 15–24% in S2 and 14–28% in S3, in both regimes. The degraded reconstructions of S2 and S3 thus add detections throughout the drive rather than near the obstacle.

Table II

R1 (SINGLE OBSTACLE): FULL PER-SCENARIO METRICS OF THE TWO DETECTORS, COMPUTED CONCURRENTLY ON IDENTICAL DATA AGAINST THE OBSTACLE LABEL. THE LABEL MEASURES SYSTEM 1'S INTENDED TASK DIRECTLY AND PROBES SYSTEM 2'S OPERATIONAL SCOPE; THE COLUMNS CHARACTERIZE THE TWO COMPLEMENTARY SIGNALS OF THE DUAL-PROCESS ARCHITECTURE.

	S1 Static		S2 Random route		S3 Random towns	
Metric	CNN	KL	CNN	KL	CNN	KL
Accuracy	0.8716	0.7831	0.7436	0.7194	0.7969	0.3566
Bal. acc.	0.7308	0.4974	0.6798	0.5198	0.7469	0.5198
Precision	0.8325	0.1063	0.4584	0.2939	0.5879	0.2609
Recall	0.4878	0.0042	0.5604	0.1464	0.6462	0.8484
Specificity	0.9739	0.9907	0.7992	0.8933	0.8476	0.1911
FPR	0.0261	0.0093	0.2008	0.1067	0.1524	0.8089
FNR	0.5122	0.9958	0.4396	0.8536	0.3538	0.1516
F1	0.6152	0.0080	0.5043	0.1954	0.6157	0.3991
MCC	0.5724	−0.0233	0.3367	0.0524	0.4790	0.0445

Table III

R2 (RANDOM OBSTACLES): FULL PER-SCENARIO METRICS OF THE TWO DETECTORS, COMPUTED CONCURRENTLY ON IDENTICAL DATA AGAINST THE OBSTACLE LABEL, UNDER THE SAME READING AS TABLE II: THE COLUMNS CHARACTERIZE THE TWO COMPLEMENTARY SIGNALS OF THE DUAL-PROCESS ARCHITECTURE.

	S1 Static		S2 Random route		S3 Random towns	
Metric	CNN	KL	CNN	KL	CNN	KL
Accuracy	0.9626	0.9650	0.7979	0.8480	0.7598	0.2523
Bal. acc.	0.9523	0.4968	0.7624	0.5072	0.7377	0.5274
Precision	0.4193	0.0050	0.1504	0.0517	0.2727	0.1165
Recall	0.9414	0.0014	0.7234	0.1315	0.7094	0.8801
Specificity	0.9632	0.9922	0.8015	0.8828	0.7660	0.1747
FPR	0.0368	0.0078	0.1985	0.1172	0.2340	0.8253
FNR	0.0586	0.9986	0.2766	0.8685	0.2906	0.1199
F1	0.5801	0.0022	0.2490	0.0742	0.3940	0.2058
MCC	0.6146	−0.0121	0.2651	0.0094	0.3292	0.0458

D. System 2 Behavior: Distribution-Shift Detection

Scored against the obstacle label, the KL threshold responds to the scenario rather than to the obstacle, as the bottom rows of Figs. 5 and 7 show. In S1 the rule is nearly silent (false-positive rate 0.009 in R1 and 0.008 in R2) and misses almost every obstacle frame (recall 0.004 and 0.001). In S2 it fires on 11–12% of the nominal frames. In the random towns of S3 it fires on most frames (false-positive rate 0.809 and 0.825, recall 0.848 and 0.880), and specificity drops to 0.19 and 0.17. In all six conditions the obstacle-level MCC stays at or below 0.052, i.e. at chance (Fig. 8b). A caution on raw accuracy under imbalance: in R2/S2 the KL rule's accuracy (0.848) exceeds the classifier's

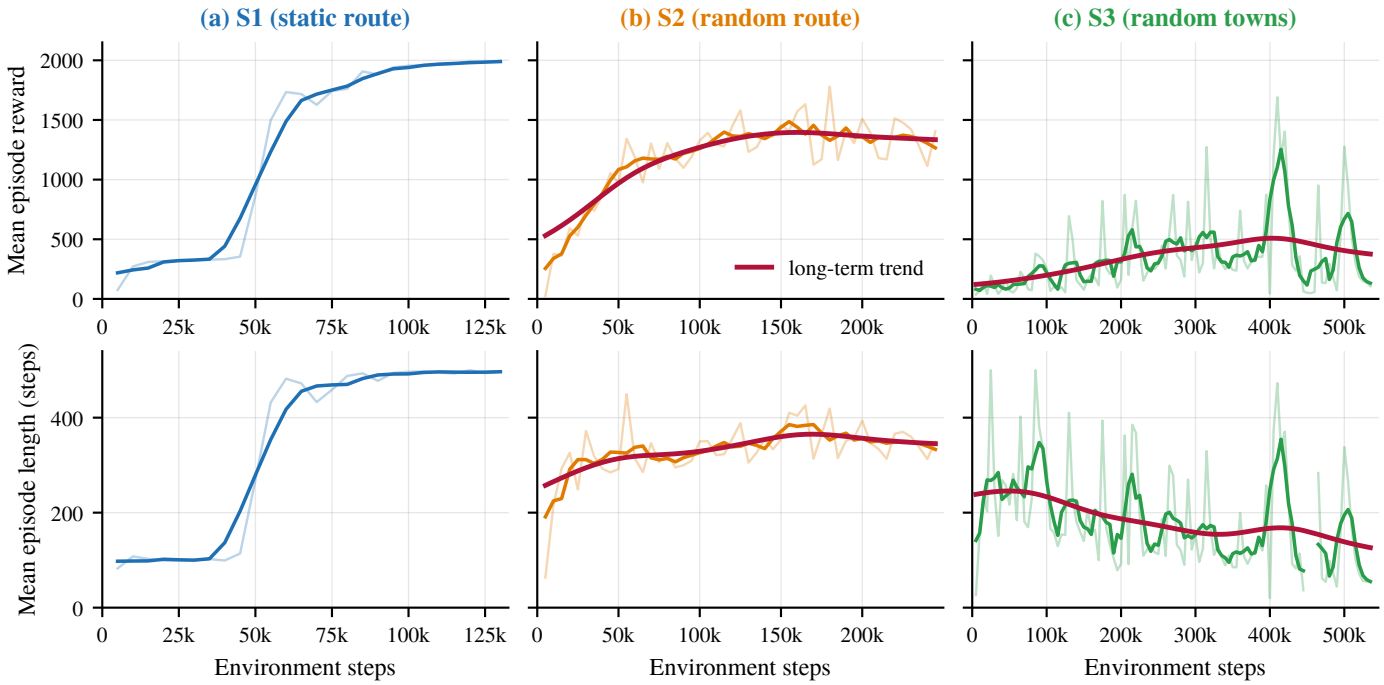


Figure 3. DreamerV3 training curves for the three curricula: mean episode reward (top) and mean episode length (bottom), one column per scenario. Faint lines are the raw logged values (one point every 5k environment steps); bold lines are a centered moving average over $N = 5$ logged points (25k steps). In S2 and S3 the red curve shows the long-term trend, a Gaussian-kernel smoothing of the raw values with a width (σ) of 10% of the run length. Each curve ends at the checkpoint used in all evaluations (S1 131k, S2 246k, S3 535k steps). (a) S1 (static route): monotone ascent to a stable plateau of ≈2000 reward, with episode length saturating at the 500-step horizon. (b) S2 (random route): convergence to a ∼1200–1500 reward band with per-route variance but no collapse. (c) S3 (random towns): a sawtooth pattern of within-town adaptation and collapse at map changes; the smoothed reward stays below 600 on about 90% of the logged steps and reaches the S2 band only briefly, around 410k steps.

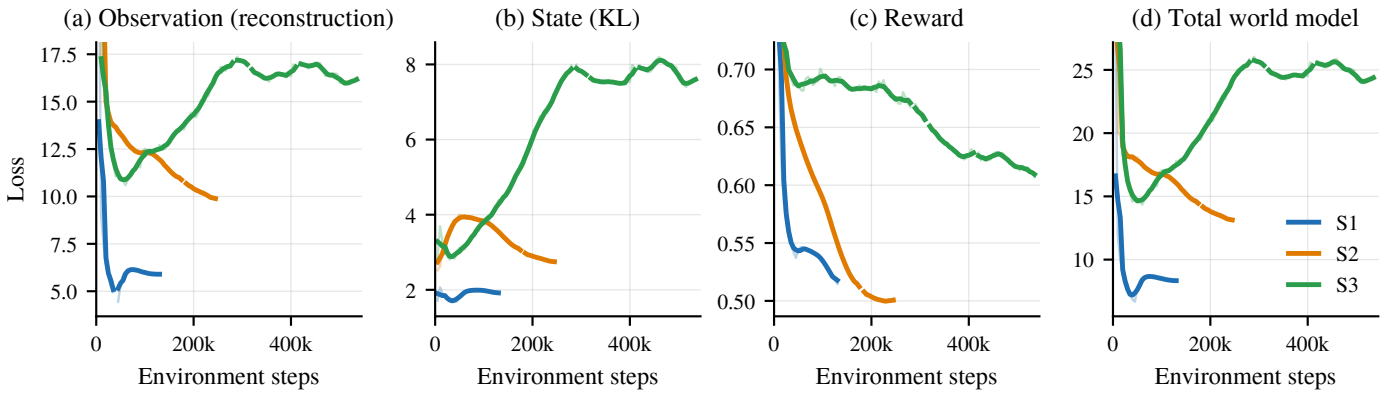


Figure 4. World-model losses during training for S1, S2 and S3, overlaid: (a) observation (reconstruction) loss, (b) state loss, the free-bits KL term $\beta_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \beta_{\text{rep}}\mathcal{L}_{\text{rep}}$ between posterior and prior, (c) reward loss, (d) total world-model loss. Faint lines are raw values and bold lines a centered moving average over $N = 5$ logged points; the y-axes start after the first 20k steps, so the large initial values are clipped. Each curve ends at the evaluated checkpoint. S1 and S2 converge, whereas in S3 the observation, state and total losses rise after an initial drop while the reward loss keeps decreasing.

(0.798), yet its balanced accuracy (0.507 vs. 0.762) and MCC (0.009 vs. 0.265) show that this figure reflects the 4.6% anomaly prevalence rather than separation of obstacle frames.

This pattern delineates the operational scope of the method and confirms that the KL threshold operates as designed: it is a *distribution-shift* detector [2, 7], and the obstacle label serves as a probe of that scope. A single static obstacle occupies a small fraction of the scene, and its contribution to a global KL over the 32×32 latent is swamped by everything else; a town change, by contrast, shifts the entire distribution and is detected immediately and aggressively, which is exactly the behavior a model-validity monitor should exhibit.

E. Analysis: Why the Operating Regimes Are Complementary

Both detectors depend on the fidelity of imagination, and reconstruction fidelity is what degrades in S3 (observation loss 11 → 17, world-model loss 15 → 25; Fig. 4). In S1 the world

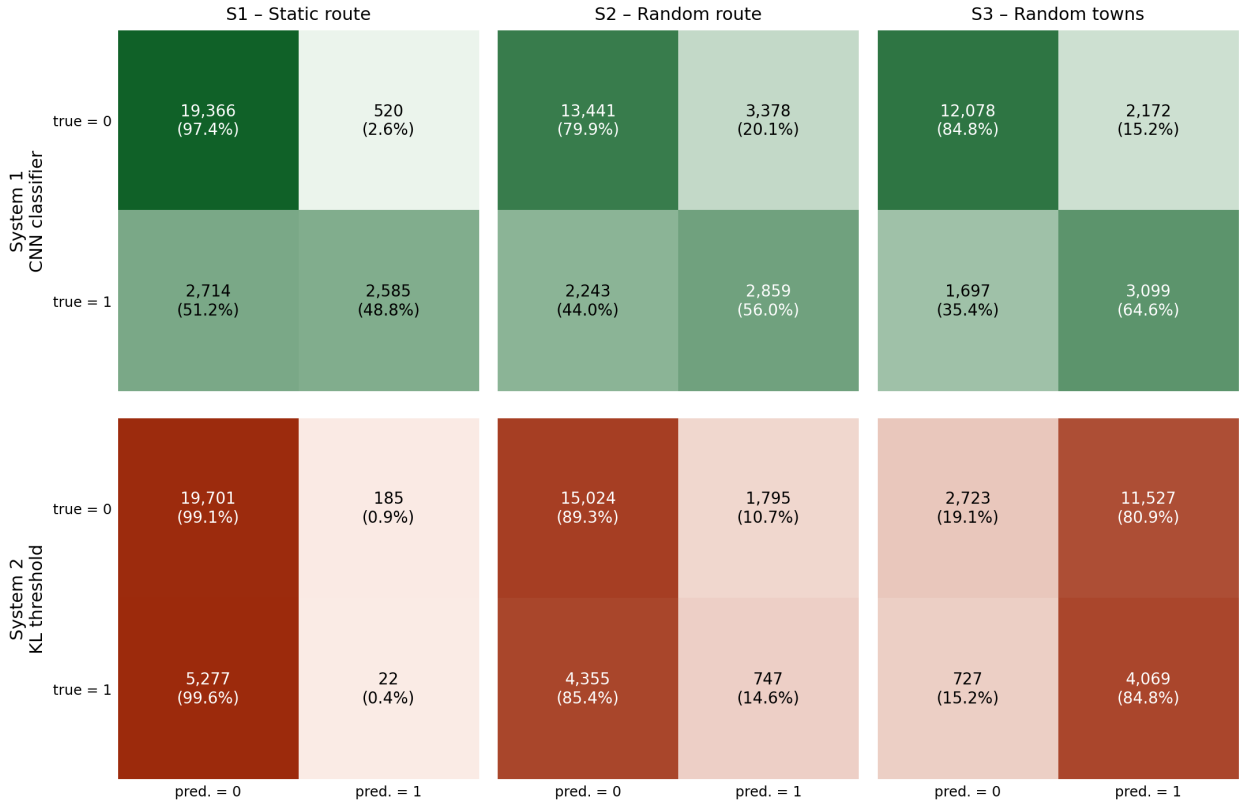


Figure 5. R1 (single obstacle): per-timestep confusion matrices, against the obstacle label, for the System 1 CNN classifier (top row) and the System 2 KL threshold (bottom row) across scenarios S1-S3. The classifier's false-positive share is smallest in S1 and grows in S2 and S3; its missed frames are mostly frames in which the class-conditional label already marks a distant car (Section V-C). The KL rule's matrices expose its two opposite response modes on this label: in S1 nearly all anomalous frames land in the false-negative cell (the rule almost never fires), whereas in S3 most nominal frames land in the false-positive cell (the rule fires on most frames), reflecting that it tracks scene familiarity.

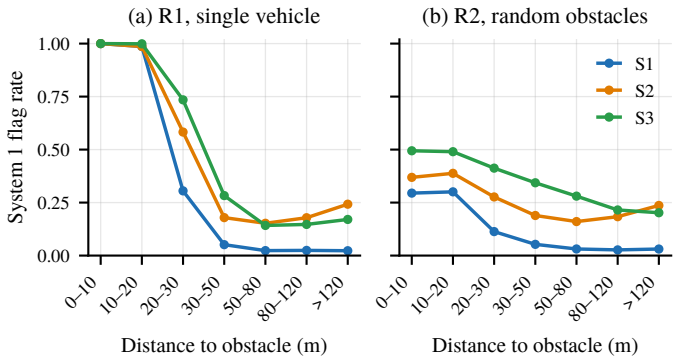


Figure 6. System 1 flag rate as a function of the distance to the obstacle, in the open-loop baseline runs (100 episodes per scenario; per-frame flags of the classifier at $\tau = 0.149$). (a) R1, single vehicle obstacle. (b) R2, random obstacles. Under the well-fitted world model of S1 the classifier fires mainly near the obstacle over the part of the route driven before it; under the degraded reconstructions of S2 and S3 it also fires on 14–28% of the frames far from it. In R2 the near-field rate is lower because most obstacles are small static props (Section VI).

model is well fitted: the classifier reaches its highest MCC and the KL rule stays silent. In S3 the reconstruction has not converged: the classifier's precision is below its S1 level and the KL rule fires on most frames. S2 lies between the two: the KL rule stays mostly silent while the classifier's precision is at its lowest, so the route-level degradation of S2 affects System 1 without raising System 2. The control-reconstruction decoupling of Section V-B (reward loss falls while reconstruction degrades) is consistent with this picture.

It suggests that reconstruction-based safety supervision can become unreliable before control fails, because obstacle detection depends directly on reconstruction fidelity whereas the learned policy may remain partially functional. The reconstruction side of this divergence is measured by the world-model losses and the detector precision; obstacle-free runs of the unsupervised agent, 100 episodes per scenario, measure the control side. The policy completes the route or reaches the episode horizon without failure in 92, 81 and 57 episodes in S1, S2 and S3, so control also degrades, but gradually, with a median route completion of at least 97.6%.

VI. A Unified Two-Tier Anomaly Observation System

A. Decision Structure

Table IV contrasts the two detectors. They answer different questions, exploit different signals, and reach their operating

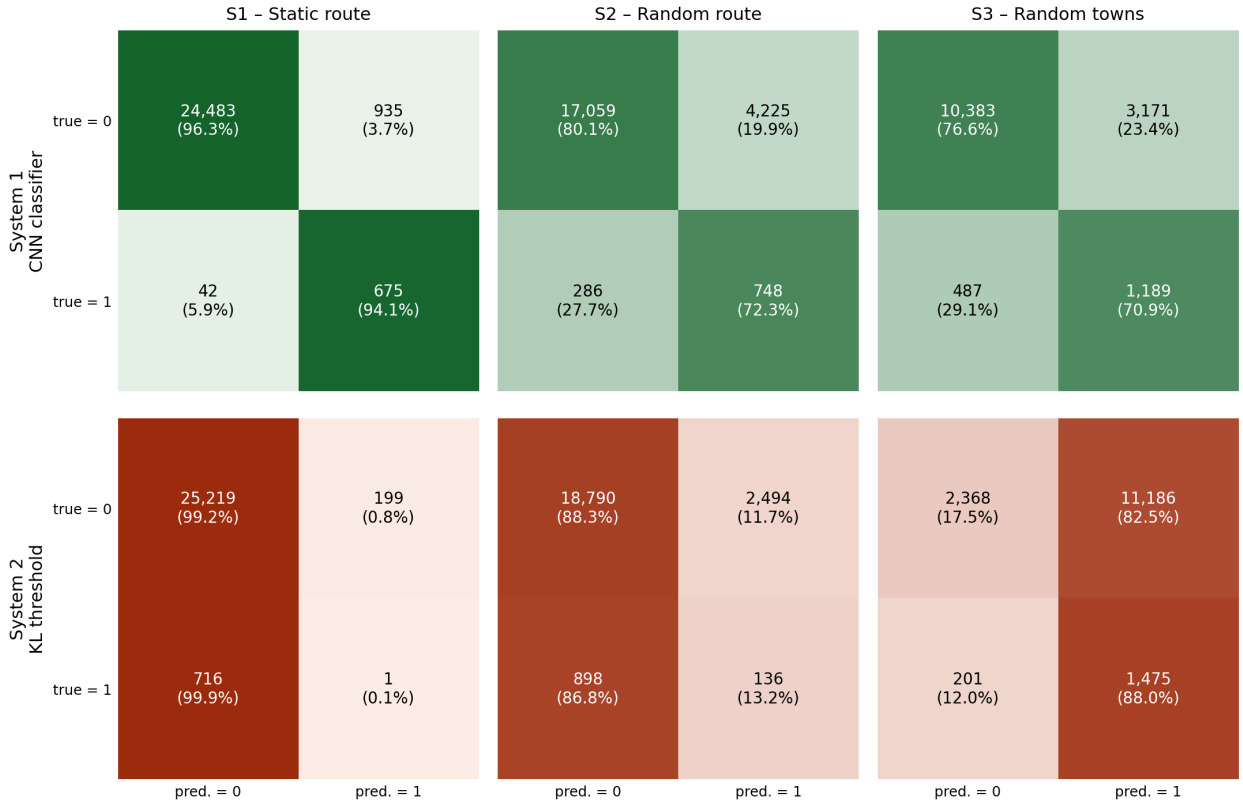


Figure 7. R2 (random obstacles): per-timestep confusion matrices, against the obstacle label, for the System 1 CNN classifier (top row) and the System 2 KL threshold (bottom row) across scenarios S1-S3. Relative to R1 (Fig. 5), the classifier's recall is higher, since the label is the class-agnostic operator the classifier is trained on, and its precision is lower, at the lowest anomaly prevalence of the study (Table I). The KL rule reproduces the pattern of R1: silent under a well-fitted world model, firing on most frames under a novel one.

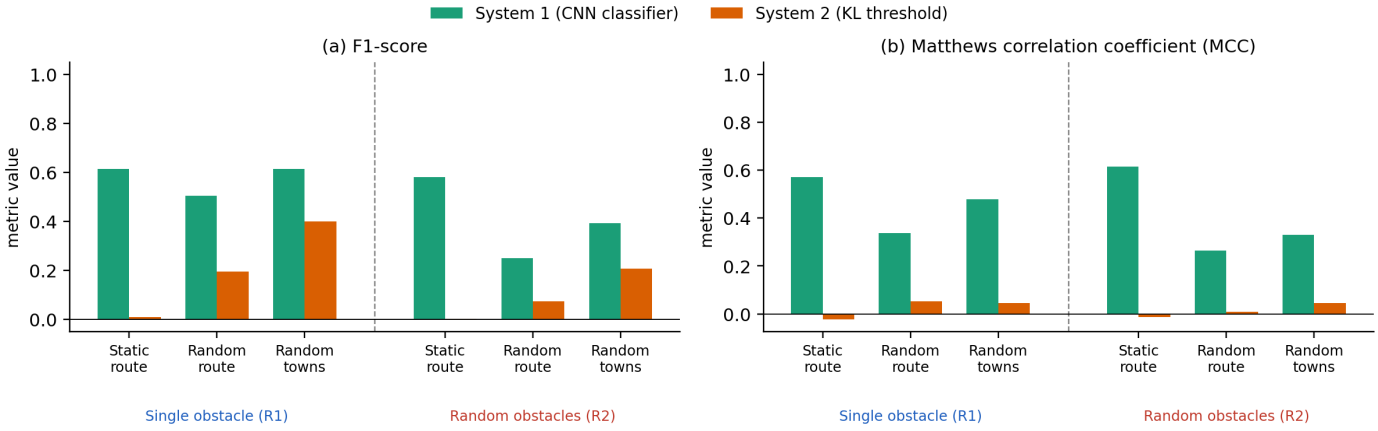


Figure 8. System 1 (CNN classifier) and System 2 (KL threshold) scored against the common obstacle label, in terms of (a) F1 and (b) MCC, across all six evaluation conditions. The classifier reaches its highest scores in S1 and lower scores in S2 and S3; the KL rule stays near chance on obstacle-level detection in every condition, consistent with its role as a distribution-shift detector. The two bars characterize the two complementary signals on the same data.

regimes under opposite world-model conditions, the defining property of complementary System 1 / System 2 processes. The measured regimes of Section V-C define a two-tier *Anomaly Observation System*, whose decision structure follows directly from Tables II–III:

- **Tier 1 (System 1, primary detector).** The CNN classifier runs at every timestep as the in-distribution obstacle detector, providing fast local obstacle detection from discrepancies between observed and reconstructed semantic frames, as measured in Section V-C.
- **Tier 2 (System 2, model-validity supervisor).** The KL monitor continuously assesses whether the agent is in an environ-

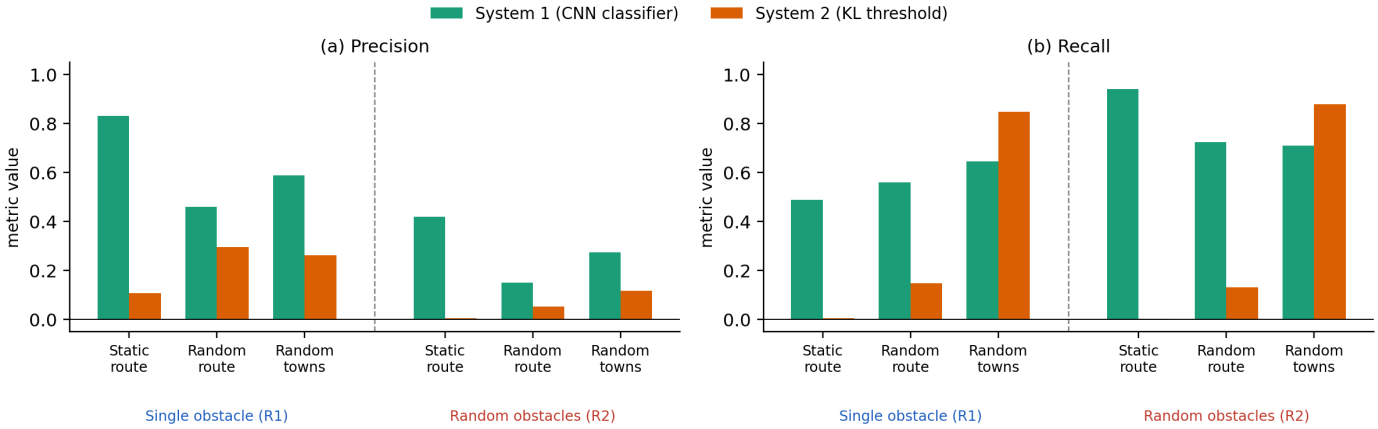


Figure 9. System 1 (CNN classifier) and System 2 (KL threshold) scored against the common obstacle label, in terms of (a) precision and (b) recall. The classifier's precision is highest in S1 in both regimes and lowest in S2. The KL rule's recall is near zero when the world model is well fitted (S1) and above 0.84 when it has broken down (S3), where it fires on most frames, the signature of a detector tracking global model validity.

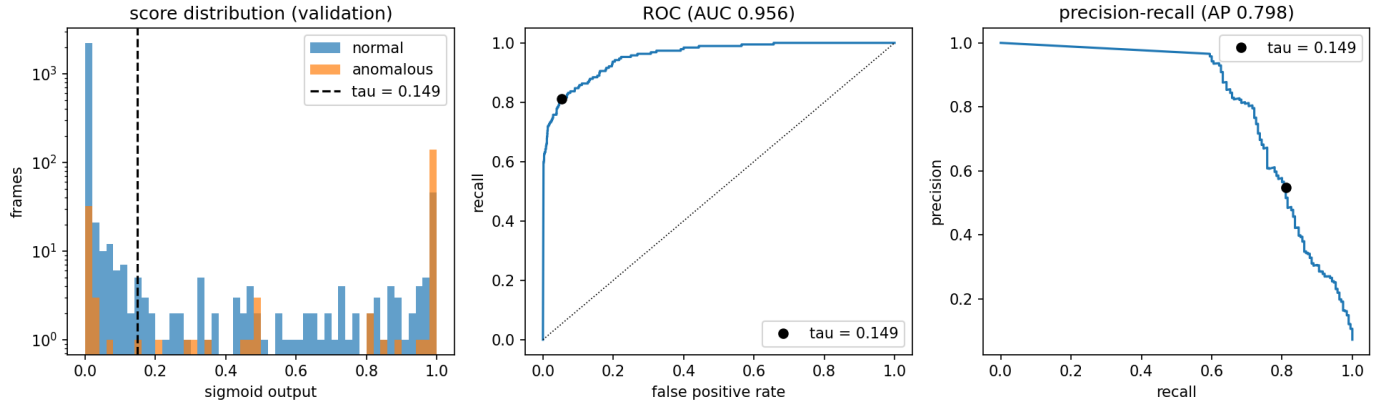


Figure 10. Selection of the System 1 decision threshold on the validation split (2,601 frames, 190 anomalous). Left: score distribution of the sigmoid output for nominal and anomalous frames (log scale), with the selected $\tau = 0.149$. Middle: receiver operating characteristic (area 0.956); the marker is the operating point of the selected threshold, recall 0.811 at a false-positive rate of 0.053. Right: precision-recall curve (average precision 0.798) with the same operating point.

Table IV
CONCEPTUAL CONTRAST BETWEEN THE TWO DETECTORS UNDER THE DUAL-PROCESS FRAMING.

Property	System 1: CNN classifier	System 2: KL threshold
Signal nature	Local, spatial	Global, distributional
Detects	Obstacle presence	Distribution shift / novelty
Question answered	"Is there an obstacle in my path?"	"Do I understand this environment?"
Best regime	Well-fitted world model (S1)	Abrupt scene change (S3)
Failure mode	More false positives under a degraded reconstruction (S2, S3)	Misses obstacles, or fires constantly
World-model dependence	Indirect (reference reconstruction only)	Direct (intrinsic to the RSSM)
Transferability	Architecture-agnostic module on image pairs	Requires explicit prior/posterior (RSSM)

ment its world model understands, the behavior measured in Section V-D. Its role is to signal when Tier 1 itself has ceased to be trustworthy: the architecture assigns a sustained high-intensity global KL signal the function of down-weighting confidence in the classifier's outputs. This assignment is grounded in the measurements: the KL signal saturates in S3, where the classifier's precision is below its S1 level. In S2 the classifier's precision is lower still while the KL signal stays mostly silent, so the gate covers the town-level shift of S3 and leaves the route-level degradation of S2 to System 1 alone.

B. *Closed-Loop Realization*

The decision structure above is implemented as a per-step supervisor (Fig. 1, bottom) whose command is a brake level in $[0, 1]$. The brake is *progressive*: a run of consecutive System 1 detections raises the brake by an increment δ per frame, up to a full stop, so that a detection that persists brings the vehicle to rest while an isolated detection produces at most a light brake. A run survives a gap of up to g frames without a detection and is reset to zero by a longer gap, which releases the brake. This temporal logic is designed to separate a real obstacle from the

reconstruction errors of an imperfect world model: an obstacle stays in the corridor and keeps the flag up frame after frame, whereas reconstruction errors tend to be less persistent and are less likely to accumulate into a full stop. The System 2 alarm is the exponential moving average of the per-frame novelty bit of (5), $r_t = \beta\, r_{t-1} + (1-\beta)\, D_t^{(2)}$, compared against a threshold ρ; the average converts the per-frame rule into a statement about *sustained* surprise and is silent on isolated firings. While the alarm is down, the ramp starts on the first System 1 detection. While the alarm is up, System 1 is down-weighted: the ramp starts only after k consecutive detections, following the measurement of Section V-C that the classifier's precision is reduced in S3, where the alarm is raised. The System 2 alarm has no other effect on the vehicle in the two-tier configuration; its role is confined to gating the trust placed in System 1. In the standalone SYSTEM 2 configuration a run of consecutive alarm frames drives the same ramp, which measures what the monitor achieves as a safety trigger on its own. Values used in this study: $\delta = 0.25$, $g = 1$, $\beta = 0.9$, $\rho = 0.5$, $k = 3$.

These parameters were selected during development runs and testing to provide a gradual braking response while suppressing isolated detector activations: $\delta = 0.25$ yields a full brake on the fourth consecutive detection, $g = 1$ tolerates a single-frame interruption, $\beta = 0.9$ smooths transient System 2 alarms, $\rho = 0.5$ requires sustained novelty before gating is enabled, and $k = 3$ delays System 1 intervention when the world model is considered unreliable. All commands are computed from the frame at time t and applied at $t+1$.

C. Safety Evaluation

Table V reports the outcomes of the closed-loop pass for both obstacle regimes and all three driving scenarios. The episodes of each configuration are drawn independently, with routes and obstacle placements sampled afresh, so configurations are compared through the 95% intervals of their collision rates.

Single obstacle. Without supervision the agent collides with the parked car in 98, 84 and 75 of 100 episodes in S1, S2 and S3. With SYSTEM 1 or SYSTEM 1 + 2 no collision occurs in any scenario. In S1, 91 and 94 episodes end in an obstacle stop, with a median stopping margin of 18.1 and 18.0 m after a brake onset at a median of 22 m from the obstacle, and early braking occupies 4.8% and 4.4% of the frames. As the reference reconstruction degrades, the avoided collisions are increasingly obtained by early stops: 73 and 71 episodes in S2, 69 and 60 in S3, with early braking on 22–25% of the frames in S2 and 16–23% in S3. This is consistent with the precision drop of Section V-C: as the reconstruction degrades, the supervisor fails conservatively, stopping early rather than colliding.

System 2 alone. In S1 the standalone monitor raises an alarm on at most 0.1% of the frames and its collision rate matches the baseline's within the intervals. In S3 it stops the vehicle early in 92 (R1) and 96 (R2) episodes, after a mean of 46 and 38 steps, on alarms raised by the unfamiliar town regardless of obstacle presence. The monitor tracks scene familiarity, which is its role as a model-validity supervisor.

Two-tier gating. In S1 and S2 the System 2 alarm is up on 0.7–1.4% and 8.8–10.3% of the frames, and SYSTEM 1 + 2 behaves as SYSTEM 1: their collision rates coincide in R1 and in R2/S2 and lie within each other's intervals in R2/S1. In S3 the alarm is up on 83.6% (R1) and 72.9% (R2) of the frames and the gate is active: in R1 early stops fall from 69 to 60 and obstacle stops rise from 18 to 28, and in R2 collisions fall from 21 to 14 ($[0.14, 0.30]$ and $[0.09, 0.22]$). These differences suggest the direction of the gate's effect under a degraded world model; however, because each configuration is evaluated on an independently sampled set of episodes and the confidence intervals overlap, they are not interpreted as evidence of statistically significant superiority.

Random obstacles by class. The collision rates of R2 separate by obstacle class. In S1, SYSTEM 1 stops for 26 of the 31 vehicle obstacles and collides with none of them, and SYSTEM 1 + 2 stops for 19 of 22 with a single collision. Static props such as helmets, instrument cases and watering cans are small against the 24×24 corridor window at the 64×64 resolution of the reference; the supervisor stops for 2 of the 69 props under SYSTEM 1, and the props account for all but one of the R2/S1 collisions of the two supervised configurations.

D. Coverage and Transferability

This division covers two orthogonal safety axes, local object presence and global model validity, that neither detector covers alone: the classifier does not "know" when its reference reconstruction is unreliable, and the KL threshold does not "see" the individual obstacle.

Their combination is designed to address two distinct failure modes, local obstacle presence and global degradation of the world model: System 1 provides the obstacle-sensitive signal, while System 2 identifies conditions in which the reconstruction used by System 1 becomes less reliable. The closed-loop results of Table V show that the benefit of this interaction is regime-dependent: the combined system behaves similarly to System 1 when the world model is reliable, while the System 2 gate becomes active primarily under the degraded S3 condition. The architecture is stated here at the level of its decision structure; its per-step input signals are exactly the quantities logged and characterized in Section V-C.

The KL monitor is inherently tied to agents with an explicit world model: without prior and posterior distributions it is undefined. The classifier, by contrast, is architecturally decoupled from the controller's internals; it operates on image pairs (real observation vs. expected scene) and could in principle accompany a model-free agent, e.g., a PPO driver [26], provided a reference representation of the expected scene is supplied by a separately trained predictive model. The System 1 tier is thus a potentially general safety-supervision mechanism, while the System 2 tier is specific to the world-model ecosystem.

Table V

CLOSED-LOOP SAFETY EVALUATION OF THE ANOMALY OBSERVATION SYSTEM, 100 INDEPENDENT EPISODES PER CONFIGURATION. RATES ARE FRACTIONS OF EPISODES; AN OBSTACLE STOP ENDS WITHIN 25 M OF THE OBSTACLE, AN EARLY STOP FARTHER FROM IT; THE REMAINING EPISODES END IN A ROUTE COMPLETION, A WAYPOINT LOSS (WP LOST), A LANE DEPARTURE OR A TIMEOUT, AND THE SEVEN OUTCOME COLUMNS SUM TO ONE. STOPPING MARGIN AND BRAKE-ONSET DISTANCE ARE MEDIANS OVER OBSTACLE STOPS, IMPACT SPEED IS THE MEAN OVER COLLISIONS, EARLY BRAKING AND ALARM ARE PERCENTAGES OF FRAMES. SYSTEM 1 + 2 IS THE SYSTEM UNDER TEST, SYSTEM 1 AND SYSTEM 2 ARE ITS ABLATIONS, BASELINE IS THE OPEN-LOOP AGENT.

Scen.	Configuration	Collision rate [95% CI]	Obst. stop	Early stop	Route compl.	WP lost	Lane dep.	Time-out	Stop margin (m)	Brake onset (m)	Impact (m/s)	Early brake (%)	S2 alarm (%)
R1, single obstacle													
S1	BASELINE	0.98 [0.93, 0.99]	0.00	0.00	0.00	0.02	0.00	0.00	n/a	n/a	7.45	0.0	0.0
	SYSTEM 2	0.98 [0.93, 0.99]	0.00	0.00	0.00	0.02	0.00	0.00	n/a	n/a	7.48	0.0	0.0
	SYSTEM 1	0.00 [0.00, 0.04]	0.91	0.08	0.00	0.01	0.00	0.00	18.1	22.2	n/a	4.8	0.0
	SYSTEM 1 + 2	0.00 [0.00, 0.04]	0.94	0.06	0.00	0.00	0.00	0.00	18.0	22.1	n/a	4.4	1.4
S2	BASELINE	0.84 [0.76, 0.90]	0.00	0.00	0.00	0.08	0.01	0.07	n/a	n/a	7.85	0.0	0.0
	SYSTEM 2	0.58 [0.48, 0.67]	0.05	0.33	0.00	0.02	0.01	0.01	23.0	25.6	7.58	3.9	4.4
	SYSTEM 1	0.00 [0.00, 0.04]	0.25	0.73	0.00	0.02	0.00	0.00	18.5	22.7	n/a	25.0	0.0
	SYSTEM 1 + 2	0.00 [0.00, 0.04]	0.28	0.71	0.00	0.00	0.00	0.01	17.6	22.0	n/a	21.9	8.8
S3	BASELINE	0.75 [0.66, 0.82]	0.00	0.00	0.00	0.20	0.05	0.00	n/a	n/a	7.60	0.0	0.0
	SYSTEM 2	0.02 [0.01, 0.07]	0.03	0.92	0.00	0.02	0.01	0.00	13.3	16.2	6.05	35.4	37.3
	SYSTEM 1	0.00 [0.00, 0.04]	0.18	0.69	0.00	0.12	0.01	0.00	18.6	22.7	n/a	23.2	0.0
	SYSTEM 1 + 2	0.00 [0.00, 0.04]	0.28	0.60	0.00	0.11	0.00	0.01	16.9	21.3	n/a	15.9	83.6
R2, random obstacles													
S1	BASELINE	0.96 [0.90, 0.98]	0.00	0.00	0.00	0.00	0.01	0.03	n/a	n/a	7.46	0.0	0.0
	SYSTEM 2	0.93 [0.86, 0.97]	0.01	0.00	0.00	0.02	0.00	0.04	5.0	2.7	7.42	0.0	0.1
	SYSTEM 1	0.64 [0.54, 0.73]	0.28	0.08	0.00	0.00	0.00	0.00	17.1	21.4	7.36	5.0	0.0
	SYSTEM 1 + 2	0.73 [0.64, 0.81]	0.20	0.06	0.00	0.00	0.00	0.01	17.5	21.0	7.20	5.5	0.7
S2	BASELINE	0.82 [0.73, 0.88]	0.00	0.00	0.02	0.09	0.00	0.07	n/a	n/a	7.75	0.0	0.0
	SYSTEM 2	0.68 [0.58, 0.76]	0.04	0.26	0.01	0.00	0.00	0.01	17.2	21.6	7.86	3.3	3.7
	SYSTEM 1	0.30 [0.22, 0.40]	0.15	0.53	0.00	0.02	0.00	0.00	17.3	20.9	7.81	19.8	0.0
	SYSTEM 1 + 2	0.30 [0.22, 0.40]	0.11	0.53	0.00	0.04	0.00	0.02	11.2	14.0	7.43	18.9	10.3
S3	BASELINE	0.56 [0.46, 0.65]	0.00	0.00	0.00	0.41	0.03	0.00	n/a	n/a	7.52	0.0	0.0
	SYSTEM 2	0.00 [0.00, 0.04]	0.00	0.96	0.00	0.04	0.00	0.00	n/a	n/a	n/a	45.1	45.1
	SYSTEM 1	0.21 [0.14, 0.30]	0.10	0.57	0.01	0.10	0.00	0.01	16.2	18.6	7.77	18.4	0.0
	SYSTEM 1 + 2	0.14 [0.09, 0.22]	0.09	0.59	0.00	0.17	0.01	0.00	17.9	21.2	7.57	14.6	72.9

VII. SCOPE AND CONSTRAINTS

The results hold under the conditions of the study, which we state here.

Static obstacles. The obstacles injected in both regimes are static. The imagined scene of a moving actor legitimately contains motion, and the discrepancy signal in that setting is a separate question from the one studied here.

Monitored corridor. The classifier operates on a fixed 24×24 window over the drivable corridor. This keeps the detector lightweight and focused on the road ahead; the reported detection figures apply to hazards within that window.

Two obstacle regimes. The System 1 detector is trained on the random-obstacle collection and evaluated within both regimes of Section V-A; the R1-vs-R2 comparison of Section V-C shows how its precision responds to obstacle variety.

S3 as a stress test. The S3 world model did not converge in reconstruction (Section V-B), so the S3 columns characterize both detectors under a degraded reference and are read as a stress test of that regime.

Obstacle size. The corridor window resolves vehicle-sized obstacles; small static props are detected in few episodes (Section VI), and the R2 collision rates of the supervised configurations are dominated by them.

Sample. Results are drawn from one training run per scenario and 100 evaluation episodes per configuration, with per-timestep sample counts in the tens of thousands (Table I). The episodes of each configuration are drawn independently, so configurations are compared through their confidence intervals.

Because one independently trained DreamerV3 model is used per scenario, differences among S1–S3 reflect both the intended training-distribution differences and the stochasticity of individual training runs. The present study therefore focuses on characterizing the observed operating regimes rather than estimating across-seed performance variability.

Frames after an intervention. Once the supervisor engages the brake, the vehicle executes a command other than the one the actor issued. The safe stop therefore truncates the episode, and the detection metrics of Section V-C are taken from the open-loop pass, in which no intervention occurs.

VIII. CONCLUSIONS

This work asked how a world-model-based driving agent should react when facing situations outside its training distribution. In the setting studied here, local obstacle detection and global distribution-shift detection behave as two distinct problems: they are addressed by different signals that reach their operating regimes under different world-model conditions, a decomposition that maps onto the System 1 / System 2 distinction

used here as a functional analogy. Concretely, the paper delivers a CARLA-Gymnasium environment and trained DreamerV3 agents; a DreamerV3-internal implementation of the KL novelty threshold, which in this study behaves as a distribution-shift detector rather than an obstacle detector; a System 1 obstacle classifier that detects obstacles in the near field of the corridor and loses precision as the reconstruction degrades; and the empirically observed complementarity between the two operating regimes, on which the two-tier architecture is built.

The experiments show that local obstacle discrepancy and global world-model novelty provide distinct information. The local classifier is most reliable when the reconstruction remains well fitted, while the KL monitor becomes active when the learned world model is globally inconsistent with the current environment. Their combination therefore provides a practical mechanism for adapting the trust placed in reconstruction-based anomaly detection. In closed-loop evaluation, the supervisor eliminates collisions with the single vehicle obstacle in all three evaluated scenarios, while the heterogeneous-obstacle experiments expose the remaining limitations associated with small objects and degraded reconstructions. These results support the proposed two-tier supervision concept while also motivating future evaluation with paired scenarios, additional training seeds, dynamic obstacles, and broader out-of-distribution conditions. The conditions under which these results hold are stated in Section VII.

REFERENCES

[1] V. S. R. Kosuru and A. K. Venkitaraman, "Advancements and challenges in achieving fully autonomous self-driving vehicles," *World Journal of Advanced Research and Reviews*, vol. 18, no. 1, pp. 161–167, 2023.

[2] D. Bogdoll, M. Nitsche, and J. M. Zöllner, "Anomaly detection in autonomous driving: A survey," *arXiv preprint arXiv:2204.07974*, 2022.

[3] J. Nitsch et al., "Out-of-distribution detection for automotive perception," *arXiv preprint arXiv:2011.01413*, 2021.

[4] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," *arXiv preprint arXiv:2301.04104*, 2024.

[5] D. Kahneman, *Thinking, Fast and Slow*. New York, NY, USA: Farrar, Straus and Giroux, 2011.

[6] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998.

[7] G. Zollicoffer, K. Eaton, J. Balloch, J. Kim, W. Zhou, R. Wright, and M. Riedl, "Novelty detection in reinforcement learning with world models," in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, 2025.

[8] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[9] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *Autonomous Robots*, vol. 42, pp. 177–196, 2018.

[10] S. Ancha, Y. Raaj, P. Hu, S. G. Narasimhan, and D. Held, "Active perception using light curtains for autonomous driving," in *Proc. Eur. Conf. Computer Vision (ECCV)*, 2020.

[11] M. R. Bachute and J. M. Subhedar, "Autonomous driving architectures: Insights of machine learning and deep learning algorithms," *Machine Learning with Applications*, vol. 6, p. 100164, 2021.

[12] A. Gupta, A. Anpalagan, L. Guan, and A. S. Khwaja, "Deep learning for object detection and scene perception in self-driving cars: Survey, challenges, and open issues," *Array*, vol. 10, p. 100057, 2021.

[13] T. Meng, J. Huang, C.-M. Chew, D. Yang, and Z. Zhong, "Configuration and design schemes of environmental sensing and vehicle computing systems for automated driving: A review," *IEEE Sensors Journal*, vol. 23, no. 14, pp. 15305–15322, 2023.

[14] M. Daily, S. Medasani, R. Behringer, and M. Trivedi, "Self-driving cars," *Computer*, vol. 50, no. 12, pp. 18–23, 2017.

[15] V. Mosin, M. Staron, Y. Tarakanov, and D. Durisic, "Comparing autoencoder-based approaches for anomaly detection in highway driving scenario images," *SN Applied Sciences*, vol. 4, p. 259, 2022.

[16] D. Gao, S. Cai, H. Zhou, H. Wang, I. Soltani, and J. Zhang, "CarDreamer: Open-source learning platform for world model based autonomous driving," *arXiv preprint arXiv:2405.09111*, 2024.

[17] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. 1st Annu. Conf. Robot Learning (CoRL)*, 2017, pp. 1–16.

[18] S. R. Richter, Z. Hayder, and V. Koltun, "Playing for benchmarks," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2232–2241.

[19] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, vol. 640, no. 8059, pp. 647–653, 2025.

[20] M. Cordts et al., "The Cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 3213–3223.

[21] G. Brockman et al., "OpenAI Gym," *arXiv preprint arXiv:1606.01540*, 2016.

[22] Farama Foundation, "Gymnasium documentation: Environment API," `https://gymnasium.farama.org/api/env/`, 2026, accessed 2026-03-20.

[23] F. Belotti et al., "SheepRL: Distributed reinforcement learning accelerated by Lightning Fabric," Eclectic-Sheep, 2023. [Software]. Available: GitHub repository.

[24] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Machine Learning (ICML)*, 2015, pp. 448–456.

[25] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[26] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.